

EMILIANO VOLPINO

Full Stack Developer — Systems Design & Backend | Node.js · TypeScript · PostgreSQL

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SUMMARY

Full Stack Developer who designs and builds complete systems, not isolated features. I take platforms from whiteboard to production and keep them running: business rules, data modelling, backend architecture, integrations, deployment and infrastructure. I currently operate four systems in production, among them a multi-tenant quality-assurance SaaS and a B2B/B2C e-commerce platform with point of sale built for a paying client. My strongest ground is what happens when a system meets reality: tenant isolation, concurrency, transactional consistency, resource limits and production failures traced to root cause. TypeScript/Node.js on PostgreSQL, with TDD. English C1, fully remote-ready.

EXPERIENCE

Full Stack Developer — Software Design & Architecture (Freelance) · Vixit Labs

2026 – Present · Buenos Aires, Argentina (remote)

- End-to-end technical ownership of custom client products: B2B/B2C e-commerce platforms, point-of-sale systems and automation engines. Sole owner of each project, from the first client meeting through deployment, monitoring and production support.
- Architectural decisions: relational data modelling, backend separation of concerns, concurrency and consistency strategy, API contract design, and infrastructure stack selection based on the client's real load and budget.
- Translating business rules into systems: I turn ambiguous operational processes —tiered pricing, minimum order rules, reconciliation, approval flows— into data models and executable, auditable logic.
- Designed “*Código Cero*”, a training programme on AI-assisted software development focused on the transition from programmer to systems designer (launching 2026). Currently delivering 1:1 technical mentoring.

Operations Analyst — Claims · Pro Insurance Solutions S.A.

February 2026 – Present

- Manage and audit 100 to 120 claims bordereaux per month, ensuring data integrity, traceability and confidentiality across internal systems.
- Detect and resolve discrepancies across high-volume record sets using advanced Excel, under strict quality metrics and internal compliance policy.
- Insurance domain knowledge from inside the claims process: operational bottlenecks, manual friction and the controls a claims-management system has to solve. I work with Operations & People to identify repetitive tasks that are candidates for automation.

E-commerce Operations and Online Sales · Self-employed

2021 – 2025

- Ran an online sales operation end to end: listings, stock, logistics and customer support. It is why I model e-commerce systems well today: I understand the business the software has to represent, not just the code.

SELECTED PROJECTS

Vixit Center — Multi-tenant quality-assurance SaaS · Vixit Labs

Next.js · TypeScript · Prisma · PostgreSQL · Auth.js v5 · Tailwind CSS · Vitest · SheetJS · Vercel

- Production platform where a contact centre audits the service quality of its campaigns: evaluations over configurable scorecards, dashboards, Excel and PDF reports, calibration between reviewers, coaching plans, and a separate portal where each agent sees their feedback and files a response.
- Multi-tenant isolation: the organisation identifier always comes from the session and never from input, and every mutation by id resolves against `{id, organisation}` before touching anything. A five-role permission matrix is enforced per section and per action.
- Scoring engine configurable per organisation —module weights, criticality, N/A handling— with a visual editor validating that weights sum to 100%. If a scorecard already has evaluations loaded it blocks structural changes and

offers to duplicate it into a new version, so the referential integrity of past records is never broken.

- Data-protection compliance implemented rather than claimed: personal-data export, anonymisation that preserves aggregate history, and an append-only activity log for audit.
- 64 automated tests passing. Support tickets are forwarded over an authenticated webhook to Vixit Platform —the management platform I run the agency on— segmented by client.

Vacuum Engine — B2B/B2C commercial automation engine · Vixit Labs

Node.js · TypeScript · Express · Prisma · PostgreSQL (Supabase) · Baileys 7 (WebSocket) · Puppeteer · Angular · GitHub Actions · Oracle Cloud VPS (Ubuntu, nginx, systemd) · Vercel

- Production platform that centralises an entire commercial operation: intercepts supplier offers across WhatsApp channels and groups, syncs four supplier catalogues daily (3,400+ products with price, stock and image) and lets any offer be redistributed from a single web inbox. Replaces roughly 3 hours of daily manual monitoring, copying and forwarding.
- Rewrote the WhatsApp integration, migrating from whatsapp-web.js + headless Chrome to Baileys 7 over WebSocket and removing the browser from the server entirely: free RAM went from ~55 MB to ~380 MB on a 1 GB VPS.
- Traced a recurring production authentication failure to root cause: terminating the process without closing the socket left the session's cryptographic state (Signal protocol ratchet) half-written, so WhatsApp rejected reconnection with a 401 and forced manual re-pairing. Fixed with a SIGTERM/SIGINT handler that closes the socket and drains pending writes before exit.
- Redesigned the ingestion architecture, moving scraping out of the backend into scheduled GitHub Actions: Chrome left the server and data extraction became decoupled from the online service. This included diagnosing why the runners could not reach the database —Supabase's direct connection is IPv6-only— and routing it through the pooler. The Puppeteer scrapers handle infinite scroll, lazy loading and retries: tuning resource interception fixed an extraction truncating at 225 products and brought it back to the full ~1,638-product catalogue.
- Performance work with a measured result: full application load from 5-6 s down to 1.1 s. In-memory caching with a stale-while-revalidate pattern and warm-up on boot, a debounced refresh collapsing a burst of 20 messages into a single query, selective column projection, nginx gzip (1.12 MB to 253 KB) and, on the Angular frontend, shareReplay caching, trackBy and removal of a getter that re-filtered 3,346 products on every change-detection cycle.
- Optimistic send endpoint that confirms in 0.48 s and dispatches in the background, with a timeout on every external call and automatic degradation to plain text if image delivery fails.
- Operations and resilience: systemd service with automatic restart and OOM-killer priority, zram at 3.3:1 compression to survive the memory ceiling, HTTPS via nginx and Let's Encrypt, hourly purge of data older than 48 h, and a health check every 30 minutes that alerts by e-mail if the backend or the WhatsApp session goes down.

B2B/B2C E-commerce Platform + Point of Sale — Distribuidora Harry's · client · distribuidoraharrys.com.ar

React (Vite) · Node.js · Express · PostgreSQL (raw SQL) · Tailwind CSS · Mercado Pago API · Resend · node-cron · Vercel / Railway / Cloudflare

- Dynamic role architecture: designed a model serving differentiated prices, minimum-order rules and discount logic for retail and wholesale customers over a single product catalogue, avoiding product duplication and the consistency problems that come with it.
- In-store point of sale synchronised in real time with web inventory, including split-payment support: one stock pool hit concurrently from the counter and the web without drift.
- Concurrency control: designed a bulk catalogue-ingestion pipeline (Excel import) hardened against race conditions, combining a reactive in-memory cache, row-level transactional locks (FOR UPDATE) and idempotent inserts (ON CONFLICT DO NOTHING). Sized for the client's ~1,700 SKU catalogue.
- Transactional hardening: no monetary value is trusted from the client. The backend independently revalidates and recalculates prices, stock and coupons before confirming each operation, closing the frontend tampering surface.
- Payment webhooks with cryptographic signature verification, automated financial e-mail reports via node-cron, and production deployment with environment separation: frontend on Vercel, backend and database on Railway, DNS and protection on Cloudflare. The site is public and live.

El Cisma — Survival Horror Game Engine · github.com/Wolpi066/ElCisma

C++ · Raylib · Tau (testing framework)

- Custom game loop and manual memory management in C++ without garbage collection, with explicit control over every entity's lifecycle and the frame budget.
- Finite state machines driving enemy AI, with transitions triggered by proximity and sound cues, and a TDD unit test suite covering collision detection and inventory logic.

Also: **Breath Shop** — e-commerce with an Angular 17+ SPA on top of an MVC backend written from scratch in vanilla PHP, with JWT and oversell protection (github.com/Wolpi066/Breath).

TECHNICAL SKILLS

Backend and data: Node.js, TypeScript, Express.js, PostgreSQL (raw SQL and Prisma), Supabase, MySQL, relational schema design, concurrency and transaction control, REST APIs, JWT, Webhooks

Frontend: React.js (Vite), Angular 17+ (Signals, standalone components, RxJS), JavaScript ES6+, Tailwind CSS, Bootstrap

Architecture and practices: Transactional systems design, MVC architecture, business-rule modelling, caching strategies and performance optimisation, TDD, third-party API and payment-gateway integration, process automation, Scrum

Infrastructure and DevOps: Linux (Oracle Cloud VPS, Ubuntu), nginx, systemd, Let's Encrypt, GitHub Actions (CI and scheduled jobs), Vercel, Railway, Cloudflare, Git/GitHub, Postman, monitoring and alerting, web scraping with Puppeteer

Other languages: C++, PHP, Java, Python, SQL · **AI applied to development:** LLMs and prompt engineering as tools for design, review and acceleration of the development cycle

EDUCATION

Higher Technical Degree in Programming — UCES · In progress, expected December 2026

Systems Analyst — Instituto Tecnológico ORT · 2020 – 2022 · **Courses:** Cybersecurity and Ethical Hacking (CoderHouse, 2025), Web Development and JavaScript (CoderHouse, 2022)

Languages: Spanish (native) · English (C1 – Advanced, written and conversational)